Procedural World Generation Using Natural Language Processing

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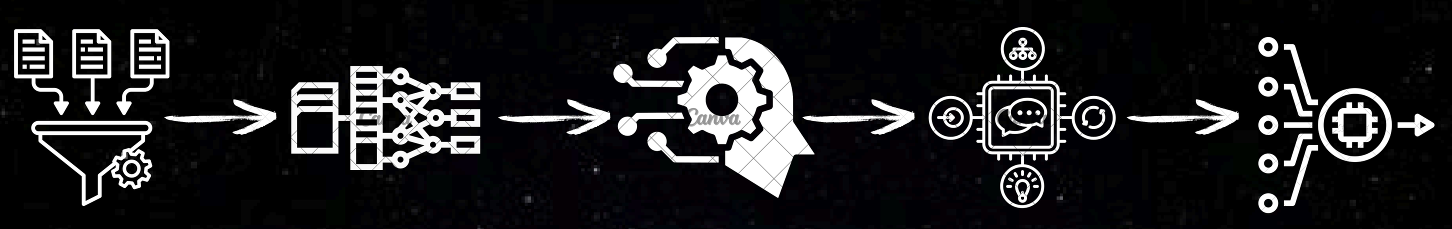
INTRODUCTION

Procedural generation creates diverse digital worlds but requires technical knowledge to achieve specific results. Our project solves this problem by letting users describe planets in plain language and automatically generating matching worlds. Instead of adjusting technical parameters, users can simply type "high oceans, low landmasses" and get exactly what they described. This approach makes procedural generation accessible to non-technical users while maintaining the benefits of algorithmic diversity.

Basic NLP Structure

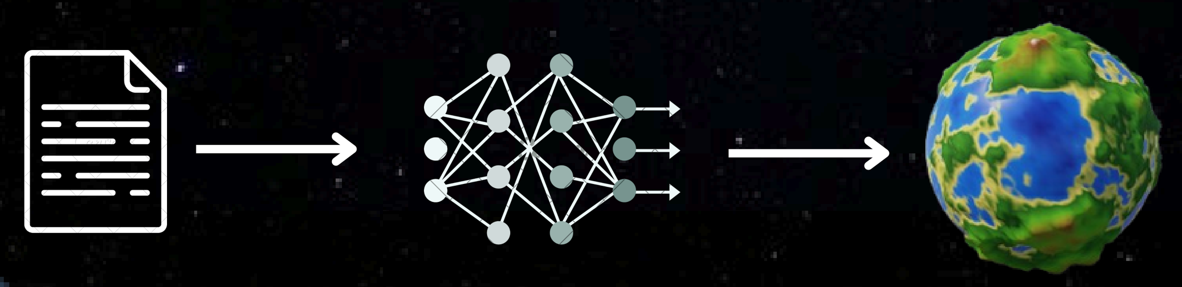


Methodology

Our system works through a straightforward process. First, we convert text descriptions into numerical vectors using language models (BERT, Gemini). These vectors capture the meaning of what the user wants. We then match these vectors with our database of parameter sets or generate new parameters using machine learning techniques (KNN, autoencoders).

The parameters control an 8-layer noise system that creates the planet's features. We use common procedural noise functions (Simplex, Perlin) to generate natural-looking terrain with the characteristics described by the user. The final planet is visualized in Unity, showing the 3D world that matches the original text description.

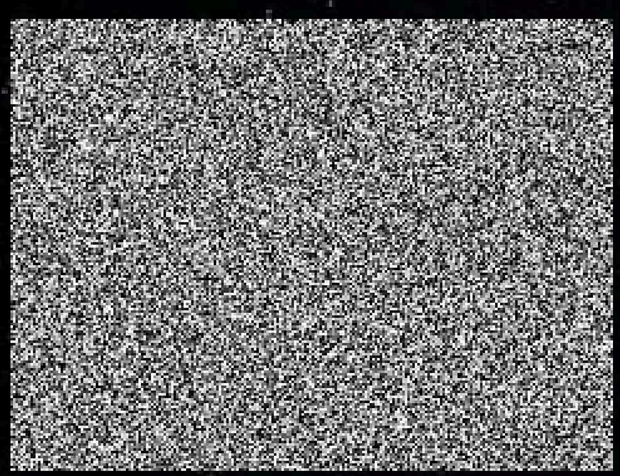
We've created a web interface where users can type descriptions, see the generated parameters, and view their planet. Our current dataset includes hundreds of text-parameter pairs, which we plan to expand to improve the system's understanding of diverse descriptions.



Noise

Procedural noise functions create natural-looking patterns that form our planet terrain. Unlike random static, these mathematical algorithms (Simplex, Perlin) generate smooth, controllable patterns. Our system uses 8 layers of noise at different scales—lower frequencies create major landmasses while higher frequencies add detailed features.

Key parameters control the terrain's appearance: strength determines feature intensity, roughness affects terrain irregularity, persistence controls detail distribution, and minValue sets ocean-to-land ratio. By adjusting these values based on text descriptions, we transform simple words into complex planetary surfaces without requiring users to understand the underlying mathematics.



White noise as an example of random noise
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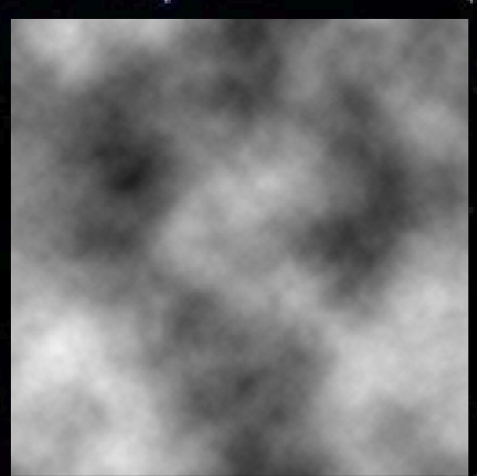
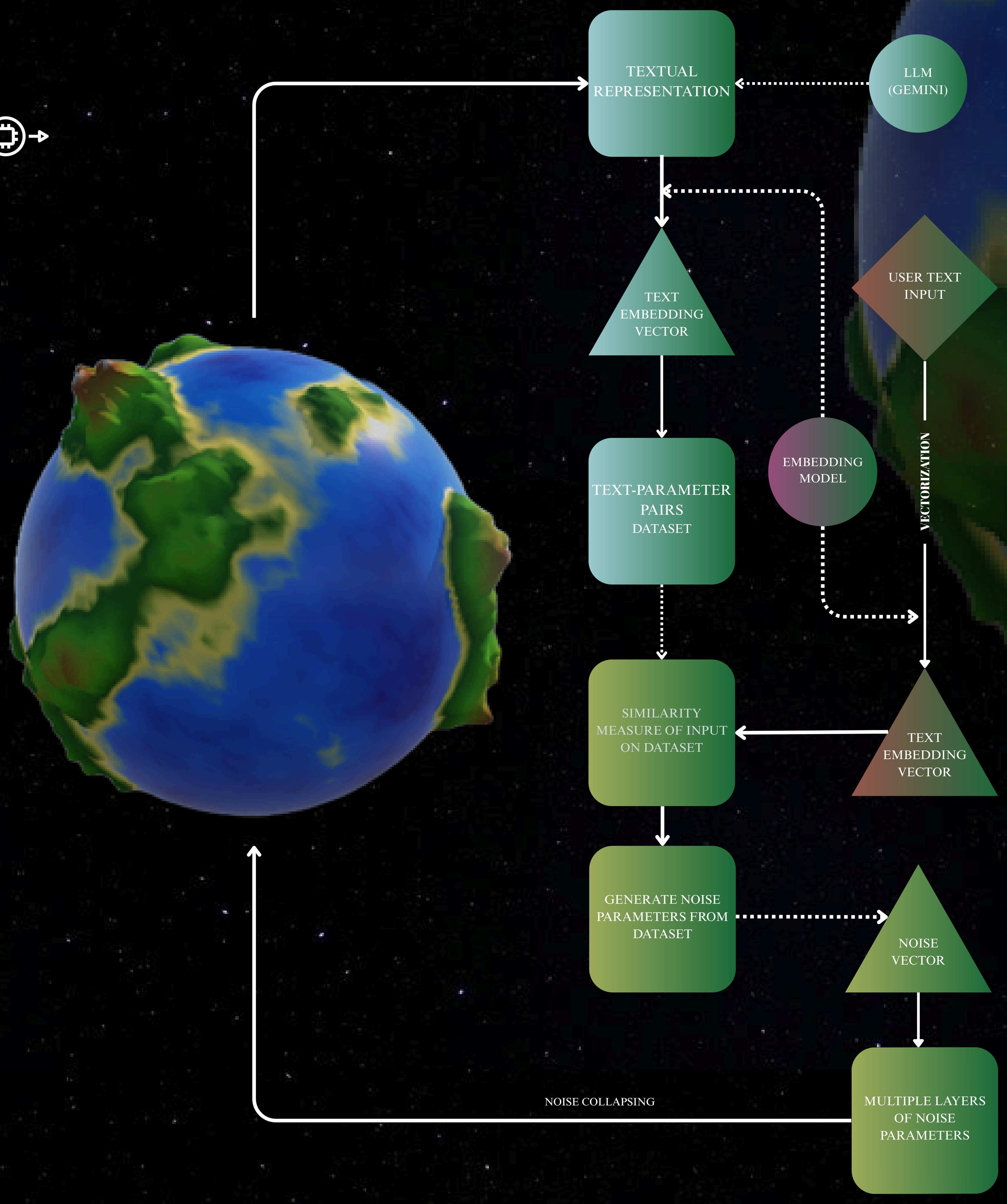


Coherent noise

Results

Our proof-of-concept demonstrates connecting text descriptions to procedural planet parameters. Users can input simple descriptions which the system matches to generate terrain visualizations. Though still in development, the prototype effectively translates basic terrain concepts into visual representations.

The system currently handles basic terrain features through our text-to-parameter pipeline, showing that natural language can indeed control procedural generation systems. Our testing confirms that even with a limited dataset, the system can match common terrain descriptions to appropriate parameter sets.



Conclusion

Our project bridges the gap between natural language and procedural generation, making technical systems accessible through everyday communication. By removing technical barriers, we enable creative professionals to harness procedural algorithms without specialized knowledge.

Potential applications include game development, simulation training, education, and virtual environments, each benefiting from simplified procedural content creation. Future work will expand our dataset, improve matching accuracy, and explore advanced vector generation techniques. This approach not only streamlines content creation but opens new possibilities for collaboration between technical and non-technical team members across industries.